SING-YU CHEN

AI Tech Lead | Full-Stack Engineer

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LinkedIn | Personal Portfolio Website

PROFESSIONAL SUMMARY

AI Team Tech Lead delivering enterprise AI systems from architecture and full-stack development through model integration and production deployment. Built 100+ AI PoCs, independently delivered 30+ projects, and contributed end-to-end engineering to 3 major government projects. Experienced in RAG, AI agents, OCR/LLM pipelines, hybrid AI and reliability engineering.

TECHNICAL SKILLS

Architecture and full-stack: Python · TypeScript · JavaScript · SQL · React · FastAPI · Node.js · REST APIs · SSE · Background Workers
AI and LLM engineering: RAG · AI agents · OCR · Prompt Engineering · Structured Output · Pydantic · Human-in-the-loop
Data and retrieval: PostgreSQL · pgvector · Supabase · DuckDB · Vector Search · BM25 · RRF · Cross-Encoder Reranking
Cloud and delivery: GCP · AWS · Docker · Cloudflare · Caddy · Testing · Monitoring · Fallback
Computer vision: OpenCV · SIFT · FLANN · affine transformation · NumPy · video processing

PROFESSIONAL EXPERIENCE

REAS Intelligence Co., Ltd. | Data Engineer / AI Team Tech Lead **Jun 2024–Present**

- Lead AI/data engineering planning, architecture and delivery; translate product requirements into testable designs and coordinate implementation across functions.
- Build AI agents, document/data pipelines, ETL, inference services and REST APIs with schema validation, fallbacks, recovery and operational monitoring.
- Evaluate model quality, latency, privacy and cloud/on-premise cost; deploy and maintain services on GCP, Docker and Cloudflare.
- Delivery record: 100+ AI PoCs, 30+ independently delivered projects, and 3 major government projects.

Revelis Biotechnology Co., Ltd. | Data Analytics Engineer Intern **Mar–Jun 2023**

- Performed data cleansing, statistical analysis, visualization and predictive-model validation with Python and R.

SELECTED ENGINEERING PROJECTS

Document Intelligence and Review Platform

Architecture: React client → FastAPI API / background workers → native parsing / OCR fallback / rules / LLM → Pydantic validation → human review → PostgreSQL
Skills: React · TypeScript · Python · FastAPI · PostgreSQL · OCR · Claude · Gemini · Docker · Pydantic

- Delivered the full stack independently; similarity prefiltering and batched inference reduced first-pass LLM calls from 36 to 1 and end-to-end time by 44.2%.

Operations and Hybrid AI Platform

Architecture: React / PWA → Node.js API → Supabase / PostgreSQL / Auth / Storage → GCP; cloud STT and on-premise Ollama / Gemma stream through SSE
Skills: React · JavaScript · Node.js · Supabase · GCP · Vertex AI · Ollama · Gemma · STT · SSE · Caddy

- Delivered public, member and operations surfaces; added hierarchical summarization, schema validation, timeouts and fallbacks. All 81 automated tests passed.

Enterprise Private AI Agent and RAG Platform

Architecture: FastAPI → PostgreSQL / pgvector + BM25 / RRF / reranking → model routing; structured intent → Pydantic → deterministic DuckDB analysis
Skills: Python · FastAPI · React · TypeScript · pgvector · BM25 · RRF · Cross-Encoder · DuckDB · Ollama · Docker

- Implemented traceable hybrid retrieval, grounded answers and tool workflows while separating language understanding from reproducible computation.

BraveTalk Club | Personal Side Project

Architecture: PWA → Cloudflare Workers → D1 / Static Assets → Workers AI; sessions, AI quotas and fallback preserve core learning and collaboration
Skills: JavaScript · Cloudflare Workers · D1 · Workers AI · Gemma · PWA · Web Speech API · MediaRecorder · Service Worker

- Independently designed, built and deployed an AI speaking and two-person travel product with 76 scenarios, 304 phrases and 7/7 automated tests.

EDUCATION AND CERTIFICATIONS

Rennes School of Business | Master in Data & Business Analytics | 2023–2025
College of Management, Fu Jen Catholic University | Department of Statistics and Information Science · B.S. | 2019–2023
Certifications: AWS Generative AI Application Development · TQC+ Professional-Level Python 3 · iPAS Junior Big Data Analyst